

Execution and Site Supervision

Methodologies for Utility-Scale Solar PV Subcontracted Projects

1. Construction Management Overview

The execution of a 100 MW utility-scale solar photovoltaic (PV) plant, coupled with a 33/220kV Grid Substation (GSS), represents a massive logistical and technical undertaking. When operating under a fully subcontracted Engineering, Procurement, and Construction (EPC) model, the primary contractor's construction team shifts from direct labor management to high-level orchestration, risk mitigation, and interface control. In this environment, the Construction Manager (RAMPART) and the discipline in-charges (BASTION, STRATUM, CONDUIT, ARCLINE, FORTRESS, SWITCHMAN, and PRISM) function as the governing authority over specialized trade subcontractors.

Supervision in a fully subcontracted model is defined by the continuous validation of the subcontractor's compliance with approved engineering designs, Method Statements (MS), Quality Assurance/Quality Control (QA/QC) plans, and commercial schedules. The EPC supervisory team does not instruct the subcontractor's labor force on how to swing a hammer or pull a cable; rather, they manage the project through formal commercial and technical mechanisms. These include the issuance of Permits to Work (PTW), the approval of Inspection Calls (ICs), the execution of Stop Work Authority (SWA), and the certification of Joint Measurement Certificates (JMCs) for billing.

The authority and responsibility matrix delineates the transfer of execution risk. The EPC supervision team dictates the sequence of work, authorizes work fronts, and sets the quality baseline. The subcontractor retains full responsibility for the provision of labor, plant, equipment, and the specific execution methodologies required to meet the EPC's standards.

Functional Area	EPC Supervision Team (RAMPART & In-charges)	Subcontractor Site Management
Site Access & Sequencing	Dictates priority; issues PTWs; de-conflicts spatial interfaces.	Requests access; adheres to approved master schedule.
Execution Methodology	Reviews and approves Method Statements; identifies safety gaps.	Authors Method Statements; selects tools and machinery.
Quality Control	Witnesses hold points; signs Inspection Calls;	Performs the work; conducts internal QA; raises

	issues NCRs.	Inspection Calls.
Progress & Billing	Verifies quantities in the field; certifies JMCs; tracks S-Curve.	Submits Daily Progress Reports (DPR); prepares billing drafts.
Safety & Environment	Audits site conditions; enforces SWA for imminent danger.	Provides PPE; conducts toolbox talks; executes hazard controls.

The Construction Manager's daily rhythm is highly structured to enforce discipline across multiple subcontracted entities. The day begins at 07:00 with a Morning Site Meeting, a brief huddle with all Subcontractor Site Managers to align on high-risk activities, heavy lifting operations, and daily production targets. Following this, the PTW issuance process validates that all hazard controls are in place for trenching, confined space entry, or energized work. By 08:30, RAMPART and the discipline in-charges conduct a comprehensive site walk. This front-line verification ensures that field conditions match the morning reports and that environmental controls—such as dust suppression, which is particularly critical in the loess soils of Uzbekistan—are actively maintained¹. At 13:00, interface de-confliction meetings are held to mediate spatial clashes, ensuring, for example, that MV cable trenching does not isolate a block where the mechanical subcontractor requires access for telehandlers. The day concludes at 17:00 with the review of Daily Progress Reports (DPRs), where the CM analyzes quantities claimed by subcontractors against actual surveyor verifications.

Critical interfaces between the civil, mechanical, and electrical subcontractors are managed via formal Handover Certificates. Surveyors must hand over verified pile peg-out positions to the civil subcontractor. The civil subcontractor must deliver driven piles within strict tolerances before the mechanical (MMS) subcontractor assumes responsibility for the work front. Subsequently, the MMS subcontractor must complete the tracker structure and torque verification before the electrical subcontractor can mount modules and commence DC harnessing. In the substation, civil foundations and transformer oil catch-pits must cure and pass compressive strength tests before the substation E&M subcontractor can skid the primary equipment into place.

2. Civil Works Supervision

The civil scope forms the critical path of the early project phases on a 100 MW solar project. BASTION, the Civil Supervision In-charge, oversees a massive mobilization of earthmoving equipment and concrete batching operations. The sequence begins with site clearing, grubbing, and grading. In regions characterized by highly erodible loess soils, such as Uzbekistan, mass grading is minimized to prevent severe wind erosion, dust storms, and structural settling¹.

Topographic tracking systems are increasingly utilized to allow tracker structures to follow the

natural terrain, vastly reducing cut-and-fill operations³. Following grading, perimeter fencing is established to secure the site. The survey team then stakes out pile locations, cable routes, and inverter pad coordinates.

The installation of solar tracker foundations predominantly utilizes driven galvanized steel sections, such as H-beams or C-channels, or pre-stressed high-strength concrete (PHC) pipe piles⁴. The pile driving process involves GPS-guided hydraulic hammers ramming the piles into the subsoil. BASTION's team must verify the driving depth against the engineering refusal criteria, monitor the pile orientation, and ensure the top-of-pile elevation is exact. A Pile Driving Analyzer (PDA) test or a static load test is overseen on initial test piles to confirm bearing capacity and shaft friction resistance before mass piling commences⁴. If hardpan or high soil resistance is encountered, pre-drilling with an auger may be required to reach the design embedment depth without deforming the steel pile¹.

Concreting operations for the central control room and inverter station foundations demand stringent QA/QC protocols to ensure long-term structural integrity. BASTION checks subgrade compaction, reinforcement tying (verifying rebar size, spacing, and cover blocks), and formwork integrity. Prior to the pour, a Pour Card (pre-pour checklist) is signed. During the pour, supervision monitors the concrete delivery tickets to verify the mix design and the time elapsed since batching. Slump tests are conducted to ensure workability without excess water, and concrete cubes or cylinders are cast for 7-day and 28-day compressive strength testing. The road and drainage network must be constructed to withstand heavy delivery trucks transporting transformers and modules. The sequence involves subgrade scarification and compaction, followed by the laying of a sub-base (gravel or crushed stone), a base course, and final grading. Drainage systems typically require trapezoidal earth channels or concrete V-drains. Supervision verifies compaction using a Sand Cone or Nuclear Density Gauge, checks California Bearing Ratio (CBR) values, and confirms the camber and slope to ensure proper water runoff.

The earthing grid installation provides the foundation for electrical safety. Excavators trench a grid network where galvanized steel flats or bare copper conductors are laid. These conductors are connected via exothermic welding (Cadweld) to create a molecular bond that will not degrade underground. BASTION supervises the trench depth, overlap lengths, and weld quality. Prior to grid design, and subsequently to verify the installation, soil resistivity is measured using the Wenner 4-pin method. This involves placing four electrodes in a straight line at equal distances; a current is injected through the outer pins, and the voltage drop is measured across the inner pins to calculate the resistivity of the soil in ohm-meters⁵.

Progress and quality are tracked through the Civil DPR and Inspection Calls (IC). The Civil DPR captures exact daily quantities: the number of piles driven, cubic meters of concrete poured, and linear meters of trench excavated, alongside manpower and machinery histograms. An Inspection Call is a formal document raised by the subcontractor 24 hours prior to a critical hold-point, such as inspecting rebar before a concrete pour. BASTION attends the location, verifies compliance with the approved drawings, and formally signs the IC. Subcontractor billing is subsequently certified via Joint Measurement Certificates (JMC), where BASTION and the Subcontractor physically measure the completed work in the field and apply these verified

quantities to the contract's Bill of Quantities (BOQ) rates.

3. MMS / Tracker Erection Supervision

The erection of a horizontal single-axis tracker (SAT) is a highly industrialized, assembly-line process overseen by STRATUM, the MMS/Tracker In-charge. The sequence begins with the installation of spherical bearings, which are bolted to the top of the driven piles. Next, heavy-gauge square or octagonal torque tubes are hoisted and pinned into the bearings, forming the continuous rotational axis of the tracker⁷. The drive system, comprising the slew gear, motor, and bull-gear, is mounted on a central drive pillar⁷. Cross-brackets, or purlins, are then bolted to the torque tube to support the solar modules. Crucially, shock absorbers (dampers) must be installed to mitigate wind-induced galloping and torsional forces that can destroy the array during extreme weather events. Finally, the PV modules are mounted using top-clamp clips or bolted directly from underneath³.

Pile driving tolerances are strictly enforced because exceedances cause the torque tube to bind, placing excess strain on the drive motor and compromising the structural integrity of the tracker

under wind load. The allowable horizontal (East-West) deviation is typically ± 20 mm, while the North-South spacing tolerance is ± 50 mm⁷. The elevation (height) tolerance is ± 30 mm,

and the plumbness (verticality) must remain within $\pm 1^\circ$ ⁷. When these tolerances are exceeded, the subcontractor is generally required to pull and re-drive the pile, or in limited cases, engineer custom adapter brackets that require formal design approval.

Torque specifications for the bolted connections are vital to prevent catastrophic loosening under dynamic wind and tracking loads. The pre-tightening torque for main pillar assemblies ranges from 240 to 260 N.m, while other structural linkages may require 190 to 230 N.m⁷.

STRATUM visually verifies compliance via witness marks; after a calibrated torque wrench "clicks" at the specified setting, the technician applies a stripe of torque seal paste across the nut, bolt thread, and washer. Any movement of the nut breaks the seal, providing an immediate visual indicator of loosening.

Module installation requires careful handling to prevent micro-cracks in the silicon cells. Modules are unboxed strictly at their final installation location. They are mounted with a minimum clearance of 6.5 mm between frames to accommodate thermal expansion, and care is taken to ensure that the small drainage holes on the underside of the module frames are not blocked⁸.

The modules are connected in series strings. The interconnecting cables must be secured to the torque tube with a minimum bending radius of 60 mm, ensuring that no MC4 connectors are positioned in areas where water could accumulate⁸.

Inspection hold points in the MMS erection sequence include the pre-module torque verification of the main structure and the sign-off of the drive-train alignment. Common Non-Conformance Reports (NCRs) raised by STRATUM include overtightening module clamps (which can shatter the tempered glass), scratching the zinc galvanization on the torque tubes during installation (which requires immediate remediation with cold-galvanizing zinc-rich paint), and leaving tracker rows overnight out of the horizontal stow position before the dampers are fully installed, which

exposes them to severe wind risk⁹.

4. DC Electrical Supervision

The DC electrical installation acts as the circulatory system of the solar plant, transporting power from the modules to the inverters. CONDUIT, the DC Electrical In-charge, supervises a sequence that begins with string wiring, where the MC4 connectors of adjacent modules are joined in series. These string cables are harnessed and routed safely along the torque tube, avoiding sharp metallic edges. The main DC cables are then pulled through UV-resistant cable trays or direct-buried corrugated pipes, connecting the array strings to the String Combiner Boxes (SCB) and ultimately to the inverter inputs.

During the cable pulling and laying process, CONDUIT ensures strict segregation of positive and negative cables to minimize the risk of catastrophic DC arc faults. Where cables are direct-buried, they must be laid on a bed of sifted, stone-free sand to prevent sharp rocks from puncturing the insulation under the weight of the backfill.

Terminations and crimping represent the most common point of failure in the DC network. The standard applied is IEC 62930 for photovoltaic cables, which dictates rigorous mechanical and electrical performance criteria¹⁰. Supervisors must verify that the subcontractor utilizes appropriately sized and freshly calibrated crimping tools. Random pull-tests are conducted on MC4 connectors to verify their mechanical yield strength, ensuring they will not separate under thermal expansion or wind vibration¹⁰. Comprehensive crimp records must be maintained, detailing the date, operator name, tool calibration ID, and the specific string ID for every termination.

Testing the DC system requires specialized equipment and strict adherence to IEC 62446-1 standards¹¹. A basic DC string test validates the polarity, Open Circuit Voltage (V_{oc}), and Short Circuit Current (I_{sc}) of each string¹¹. These measured values are normalized for ambient temperature and compared against the module datasheet. However, the most powerful diagnostic tool is the IV Curve test. A curve tracer sweeps the string from short circuit to open circuit, plotting the full current-voltage characteristic¹¹. The shape of this curve reveals defects that spot-checks cannot detect: "steps" in the curve indicate shaded modules, mismatched panels, or shorted bypass diodes, while a "soft knee" suggests high series resistance caused by poor crimps or degraded cell interconnects¹¹. The Fill Factor ($FF = \frac{P_{max}}{V_{oc} \times I_{sc}}$) is calculated, with typical values for high-quality crystalline silicon modules ranging between 70% and 85%¹³. Finally, earthing continuity checks are performed to verify that the array frames and mounting structures possess an equipotential bond to the earth grid, ensuring the safe dissipation of fault currents¹¹.

5. MV/AC Cable Supervision

The 33kV Medium Voltage (MV) collection network aggregates power from the distributed inverter stations and transports it to the main substation. ARCLINE, the MV/AC Cable In-charge, manages a highly sensitive scope of work where microscopic errors lead to explosive dielectric failures.

The MV cable laying sequence begins with trenching to a standard depth of 1.0 to 1.2 meters. A 100 mm bedding layer of sifted, stone-free sand is laid to protect the cable jacket. Cable pulling is executed using motorized winches equipped with dynamometers and cable rollers placed at tight intervals. This ensures that the maximum pulling tension and the minimum bending radius of the heavy XLPE (cross-linked polyethylene) cable are never exceeded. Following the jointing process, the trench receives a top layer of sand, concrete protection tiles, and polymeric warning tape before final compaction of the native soil.

Cable jointing at 33kV is a delicate operation. Cold shrink accessories are overwhelmingly preferred over traditional heat shrink kits due to their dynamic sealing capabilities, ability to expand and contract with the cable's thermal cycling, and a vastly lower failure rate (as low as 0.022%)¹⁴. Heat shrink relies on technicians applying perfectly even heat with a propane torch; uneven heating causes microscopic air voids that lead to partial discharge¹⁶. ARCLINE enforces absolute cleanliness inside the jointing bays (often climate-controlled tents). The removal of the semi-conductive screen must be flawless; if a jointer leaves microscopic carbon residue on the white XLPE insulation, it creates a conductive path that leads to surface tracking and eventual dielectric breakdown¹⁶. Furthermore, failure to apply void-filling mastic properly allows moisture ingress, which causes water treeing in the insulation¹⁷.

Before the trench is fully backfilled and the system is energized, the MV network must be proven via Very Low Frequency (VLF) testing in accordance with IEC 60502-2¹⁸. A 0.1 Hz AC

test voltage is applied, typically at $3U_0$ (three times the phase-to-ground voltage) for 15 to 60 minutes²⁰. The acceptance criterion is absolute: the cable must withstand the voltage without breakdown. Partial Discharge (PD) monitoring during the VLF test is highly recommended to detect the tiny, invisible electrical sparks occurring within microscopic voids in the joints before they can fail in service¹⁶.

Documentation of the MV network is critical for future maintenance and safety. Concrete or heavy-duty polymeric cable route markers are installed at 50-meter intervals, at every change of direction, and directly over every joint. ARCLINE ensures that the survey team records the exact GPS coordinates, depth, and phase arrangement of the cable prior to backfilling, generating the final As-Built drawings required for project handover.

6. Substation Civil Supervision

The 33/220kV Grid Substation is the nerve center of the solar plant. FORTRESS, the Substation Civil In-charge, oversees a condensed, high-precision civil works program. The sequence initiates with mass grading and compaction of the substation footprint to achieve a uniform bearing capacity. Immediately following, deep trenching is executed for the earthmat installation. Excavation, formwork, and pouring of the heavy equipment foundations follow, designed to support the massive static and dynamic loads of the 220kV Step-up Transformers, Circuit Breakers, Current Transformers (CT), Potential Transformers (PT), and Isolators.

Simultaneously, reinforced concrete cable trenches with embedded unistrut channels are constructed. The control room building is erected, and massive reinforced concrete firewalls (blast-walls) are constructed between adjacent power transformers to prevent cascade failures in the event of an explosion. Finally, a layer of geotextile and crushed yard stone (gravel) is

spread across the switchyard to increase surface contact resistance, suppressing step-and-touch potentials during a ground fault.

The earthing grid (earthmat) must be installed to comply with stringent IEEE 80 standards²².

Copper or galvanized steel flats are laid in a dense crosshatch pattern beneath the entire substation footprint. FORTRESS supervises the exothermic welding of all intersections, ensuring the molten copper creates a permanent molecular bond that will not corrode or loosen over decades of service. Once complete, the system is tested using the Fall-of-Potential method

to ensure the total earth resistance is well below the typical limit of 1Ω ²².

Substation cable trenches must be constructed with an appropriate longitudinal slope (e.g., 1:200) leading to sumps equipped with automatic pumps for water drainage. Supervision ensures that PVC water-stops are perfectly embedded in all construction joints to prevent groundwater ingress. For equipment foundations, the most critical supervision check is the positioning of the Holding Down (HD) bolts. FORTRESS mandates the use of rigid steel templates, often provided by the equipment manufacturer, to hold the bolts during the concrete pour, ensuring they set with millimeter precision to match the baseplates of the incoming high-voltage switchgear.

7. Substation E&M Supervision — Most Critical Role

The erection and energization of the 220kV substation carry the highest safety and financial risks of the entire EPC scope. SWITCHMAN, the Substation E&M In-charge, orchestrates a sequence that transitions the site from a civil construction zone into a live high-voltage facility. The sequence begins with the erection of galvanized steel gantries and support structures. Heavy equipment placement follows, which involves skidding the main power transformers onto their plinths and positioning the Gas Insulated Switchgear (GIS) or Air Insulated Switchgear (AIS) modules. High voltage connections are then made using rigid aluminum tubular busbars or flexible stranded conductors. Secondary systems are installed by pulling and terminating thousands of control, protection, and SCADA cables between the switchyard and the control room. Pre-commissioning testing validates every component before the final, critical step: system energization.

The installation of a 220kV power transformer is a meticulously planned operation. Skidding the massive main tank (often weighing over 100 tons) requires hydraulic skidding systems and extreme care to avoid internal shock²⁴. Once positioned, the transformer, which arrives filled with dry nitrogen to prevent moisture ingress during transit, must undergo deep vacuum drying. Vacuum pumps draw the tank down to an extreme negative pressure, aiming for a dew point of

-30°C to extract any residual moisture from the cellulose paper insulation surrounding the copper windings²⁵. High-grade mineral oil is then introduced into the tank while it remains under vacuum, preventing the formation of microscopic air bubbles that could cause partial discharge.

The transformer oil is subjected to rigorous testing. Breakdown Voltage (BDV) testing is executed per IEC 60156. The oil sample is placed in a test cell with a 2.5 mm electrode gap, and the voltage is ramped up at 2kV/s until an arc breaches the oil. The minimum acceptable

BDV for new, filtered oil is > 70 kV²⁶. A baseline Dissolved Gas Analysis (DGA) is also

performed in accordance with IEC 60599 and IEEE C57.104²⁸. DGA measures the parts-per-million (ppm) of gases like Hydrogen, Methane, Ethylene, and Acetylene; the presence of specific gases indicates thermal overheating or electrical arcing within the tank²⁸. Electrically, a Sweep Frequency Response Analysis (SFRA) per IEC 60076-18 is performed to verify that the core and windings suffered no mechanical deformation or displacement during transport³¹.

If the substation utilizes Gas Insulated Switchgear (GIS), the erection demands a clinical, dust-free environment. Sulfur Hexafluoride (SF_6) gas handling is strictly governed by IEC 62271-4³³. The gas must meet technical grade requirements as per IEC 60376, possessing a purity $> 99.8\%$ and a moisture content (dew point) of < 200 ppmv³⁴. The flange connections between GIS bays require extreme precision to avoid shearing the delicate internal conductors or damaging the O-ring seals. Once sealed, the bay is evacuated, filled with SF_6 , and checked for micro-leaks using specialized gas sniffers.

The pre-commissioning checklist for substation equipment is exhaustive. It includes Current Transformer (CT) and Potential Transformer (PT) ratio, polarity, and magnetization curve tests to ensure accurate metering and protection. Circuit Breakers undergo contact resistance (micro-ohm) and timing tests to ensure the three phases open and close simultaneously within milliseconds. Protection relays are subjected to secondary injection testing, where fault currents are simulated to ensure the relays correctly command the breakers to trip. Finally, Insulation Resistance (Megger) and Polarization Index (PI) tests confirm the dielectric health of all cables and windings³⁶.

The most dangerous phase of the entire project is the initial energization—the transition from a dead substation to a live high-voltage bus. The sequence is tightly controlled by the grid operator. Switching from dead-to-live always involves closing the Isolator (disconnect switch) *first* under no-load conditions, followed by closing the Circuit Breaker to take up the load³⁷.

During this phase, Lockout/Tagout (LOTO) requirements are absolute. Physical padlocks and danger tags are applied to all isolation points. SWITCHMAN maintains the master lockbox; any subcontractor working near live zones must place their personal lock on the box, ensuring the system physically cannot be energized until every individual has removed their lock and cleared the area.

8. Survey & Layout

PRISM, the Survey & Layout Engineer, ensures the physical manifestation of the digital design. The site layout process begins with establishing the primary geodetic baseline using Real-Time Kinematic (RTK) GPS bases. From these primary control points, the subcontractor's surveyors peg out the exact center points of the tracker drive pillars and boundary markers.

PRISM does not perform the subcontractor's daily layout work; rather, PRISM acts as the independent auditor. Utilizing a highly calibrated Total Station, PRISM randomly samples 10% to 15% of the subcontractor's pegged points before giving clearance for pile driving. If the sampled

points deviate beyond the engineering tolerances (e.g., ± 20 mm horizontal for tracker

pillars⁷), the entire block is rejected and must be re-surveyed by the subcontractor. An As-Built survey is conducted continuously as sections of the plant are completed. It maps the exact X, Y, and Z coordinates of every installed pile, underground cable route, and concrete foundation. This living document is critical for O&M troubleshooting and is a mandatory deliverable for achieving Mechanical Completion.

9. Construction Meetings & Daily Rhythm

The cadence of site meetings dictates the momentum of the project.

- **The Morning Site Meeting:** Held at 07:00, attended by the CM (RAMPART), EPC In-charges, Subcontractor Site Managers, and HSE Leads. The agenda covers safety observations from the previous day, weather forecasts (critical for wind stowing trackers and managing dust storms²), allocation of physical work fronts, and the identification of spatial clashes. The output is a coordinated daily work plan and signed PTWs.
- **The Weekly Construction Review:** Attended by the EPC Project Director, CM, Subcontractor Project Managers, and the Lead Planner. The agenda focuses on the Master Schedule, specifically reviewing the S-Curve to track Planned versus Actual Progress. Schedule Performance Index (SPI) and Cost Performance Index (CPI) are analyzed. Open NCRs are reviewed, material delivery lookaheads are discussed, and strategic bottlenecks are identified.
- **Interface Meetings:** Dedicated ad-hoc meetings aimed at resolving handover constraints between specific trades. For example, if the Civil Subcontractor commits to delivering Block 4 piles by Tuesday, the CM holds an interface meeting to ensure the MMS Subcontractor has their crews aligned to initiate torque tube erection in Block 4 on Wednesday. These handovers are formalized into a binding micro-schedule.

10. Real Document Templates

10.1 Daily Site Report (DSR)

Project: HELIOS 100 MW	Date: 12-Aug-2026	Weather: Clear, 38°C, Wind 12 km/h
HSE Metrics	LTI: 0	Near Miss: 1
Manpower	Civil Sub: 145	MMS Sub: 210
Key Activities	Planned Today	Achieved Today
Piles Driven (Nos)	400	385
Tracker Rows Erected	15	18

PV Modules Installed	1,200	1,150
MV Cable Laid (m)	1,500	1,500
Critical Issues	- Hardpan rock encountered in Block 6. Pre-drilling auger requested. - Dust storm expected Thursday; MMS team to engage stow mode.	
Signatures	<i>RAMPART (CM)</i>	<i>Project Director</i>

10.2 Weekly Construction Report

Week Ending: 16-Aug-2026	Week No: 24	Overall Progress: 42.5%
Schedule Variance	Planned: 45.0%	SPI: 0.94 (Behind Schedule)
Top 3 Risks	1. MV Jointing delayed due to high humidity ($> 80\%$). 2. Transformer arrival delayed at port. 3. Civil progress in Block 6 stalled due to rock.	
Quality Updates	NCRs Opened: 2	NCRs Closed: 1
Lookahead (Next 7 Days)	Commence DC string testing in Block 1. Pour control room roof slab. Skid 220kV Transformer.	

10.3 Inspection Call (IC) Sheet

IC Number: HELIOS-CIV-IC-045	Date Raised: 14-Aug-2026
Location: Inverter Station 4	Discipline: Civil Works
Activity for Inspection: Rebar tying and formwork for Inverter Raft Foundation prior to concrete pour.	
Reference Drawing: HEL-CIV-INV-004 Rev 2	Hold Point: YES
Subcontractor Sign: <i>[Signature]</i>	Time Ready: 15-Aug-2026, 09:00
EPC Inspection Result:	[X] Approved [] Approved w/ Comments [] Rejected
Comments: Cover blocks verified at 50mm. Reinforcement matches BBS. Pour authorized.	
EPC Supervisor (BASTION): <i>[Signature]</i>	Date/Time: 15-Aug-2026, 09:45

10.4 Pile Installation Record

Pile ID	Block	Northing (Y)	Easting (X)	Target Elev (Z)	Actual Elev (Z)	Dev (Z)	Dev (X/Y)	Plumbness	Pass /Fail
B4-R 12-P 1	B4	4567 123.4 5	3456 78.90	+123. 450	+123. 445	-5 mm	10 mm	0.2°	PASS
B4-R 12-P	B4	4567 123.4	3456 84.90	+123. 450	+123. 485	+35 mm	12 mm	0.4°	FAIL (NCR

2		5							)
---	--	---	--	--	--	--	--	--	---

10.5 Concreting Record (Pour Card)

Structure: Inverter Pad 3	Volume: 24 m³	Grade: M30
Pre-Pour Checks	Subgrade compacted: Yes	Formwork sealed: Yes
Concrete Delivery	Truck ID: MX-45	Time Batched: 08:15
Testing	Slump: 110 mm	Temp: 28°C
Post-Pour	Curing compound applied: Yes	Finish accepted: Yes
Signatures	Subcontractor QC	EPC Supervisor (BASTION)

10.6 Cable Test Record (DC String & MV VLF)

DC String Test (IEC 62446-1)

String ID	Mod Count	Target Voc	Actual Voc	Target Isc	Actual Isc	Polarity	Pass/Fail
INV1-C B4-S12	28	1250 V	1245 V	14.5 A	14.3 A	OK	PASS

MV Cable VLF Test (IEC 60502-2)

Circuit: INV1 to Main SS	Cable Type: 33kV 3C 300sqmm XLPE	Length: 1,450m
Test Voltage: 57kV ($3U_0$)	Frequency: 0.1 Hz	Duration: 30 Mins
Phase L1-E: PASS	Phase L2-E: PASS	Phase L3-E: PASS

PD Monitored: Yes	Leakage Current: 1.2 mA	Status: ACCEPTED
--------------------------	--------------------------------	-------------------------

10.7 Subcontractor Billing Certification Sheet (JMC)

Subcontractor: Terra Civil Ltd	Period: 01-Aug to 31-Aug	Invoice Ref: TC-008
BOQ Item	Description	UOM
CV-01	Site Clearing	Ha
CV-04	Driven Piles	Nos
CV-08	Cable Trenching	Lm
Deductions	Back-charge for NCR rectification (Ref: NCR-011)	
Net Certified Value		
Signatures	<i>Subcontractor PM</i>	<i>EPC CM (RAMPART)</i>

10.8 Non-Conformance Report (NCR)

NCR No: HELIOS-MMS-NCR-012	Date: 16-Aug-2026
Subcontractor: MMS Erectors Ltd	Discipline: Mechanical
Description of Non-Conformance: Random QA torque checks on Block 2, Row 15 drive pillars revealed 4 bolts torqued to 150 N.m (Specification requires 240-260 N.m). Torque seal was applied improperly, masking the deficiency.	
Root Cause (By Subcontractor): Calibration on torque wrench #4 had	

expired and the internal spring had fatigued.	
Corrective Action: 100% re-torque of all connections made by Crew 3 using freshly calibrated tools. Torque wrench #4 removed from site.	
EPC Verification (STRATUM): Complete re-torque witnessed. 10 random bolts pulled passed. NCR Closed.	

10.9 Method Statement Review Checklist

Method Statement: MV Cable Pulling	Subcontractor: Volt-Elect
Criteria Evaluated	Status / Comments
Step-by-step methodology logical and safe?	YES. Winch tension limits clearly defined.
Specific equipment list provided?	NO. Missing calibration certs for dynamometer.
Risk Assessment / JSA included?	YES. Mitigations for trench collapse adequate.
ITP (Inspection Test Plan) attached?	YES. Hold points for VLF testing clearly marked.
Overall Decision	APPROVED WITH COMMENTS (Submit certs before start)

11. Construction-to-Commissioning Handover

The transition from the construction phase to the commissioning phase is formalized through the Mechanical Completion (MC) process. The MC certificate is the contractual milestone verifying that a system or block has been physically constructed in accordance with the design drawings and specifications. Importantly, achieving MC does *not* mean the system is energized or operational; it simply means it is safely ready to be tested. The MC certificate is signed by the

Subcontractor, the EPC Construction Manager (RAMPART), the QA/QC Manager, and the Client's Representative.

Before MC is granted, a joint walkdown occurs to generate a Pre-Commissioning Punch List³⁹. This list categorizes outstanding defects to prioritize remediation:

- **Category A:** Critical safety or functional defects that strictly prohibit testing and energization. Examples include missing earth connections, severely damaged MV cables, or un-torqued structural bolts. Category A items must be cleared before the MC certificate is signed.
- **Category B:** Defects that do not prevent safe energization but must be rectified before final commercial handover (Provisional Acceptance). Examples include missing cable tags, minor paint touch-ups on galvanized structures, or incomplete trench backfilling in non-critical areas.
- **Category C:** Minor cosmetic issues agreed to be resolved during the defects liability period.

Once MC is achieved and all Category A items are cleared, the Construction Team officially hands the system over to the Testing & Commissioning (T&C) Team. At this juncture, the site's Lockout/Tagout (LOTO) procedures shift from construction control to commissioning control, drastically restricting physical access to the equipment. The handover package provided to the T&C team includes the Red-line As-Built drawings, completed Inspection and Test Plans (ITPs), BDV/VLF/SFRA test results, and documentation of all closed NCRs.

To evaluate overall plant quality post-commissioning, the Performance Ratio (PR) is calculated per IEC 61724-1⁴⁰. The PR compares the actual AC energy delivered to the grid against the theoretical energy the plant would produce if operating continuously at Standard Test Conditions (STC)⁴¹. For a modern utility-scale tracking plant in a harsh environment like Uzbekistan, an initial PR target of 80% to 85% is expected, with continuous O&M required to mitigate the severe degradation caused by high ambient temperatures and dust soiling⁴⁰.

12. Common Site Problems & Supervisory Response

Executing a 100 MW solar plant inevitably presents acute operational challenges that the Construction Manager must resolve to maintain the Master Schedule and quality baseline.

Subcontractor pile driving hits rock at shallow depth.*Scenario:* While driving piles in Block 6, the civil subcontractor reaches refusal at 1.2 meters, whereas the engineering design requires a 1.8-meter embedment for uplift resistance.*CM Response:* BASTION immediately enacts Stop Work Authority (SWA) on piling in that specific micro-zone. The structural engineering team is alerted. The technical solution typically involves mobilizing Down-The-Hole (DTH) hammer drills to pre-drill undersized holes into the rock, driving the steel pile into the hole, and backfilling the annulus with lean concrete. If the rock is too dense, the design team may issue a revision to switch to concrete micro-pile foundations or above-ground ballast blocks for that specific row.

Subcontractor deviates from approved method statement.*Scenario:* ARCLINE observes the electrical subcontractor pulling 33kV MV cable through a trench using an excavator bucket rather than a calibrated tension winch.*CM Response:* Immediate SWA is declared. The mechanical stress exerted by an excavator easily exceeds the cable's tensile limits, stretching

the copper core and irreversibly damaging the XLPE insulation and semi-conductive layers. An NCR is issued immediately. The entire pulled section of the cable is quarantined and must undergo an extended VLF test with rigorous Partial Discharge (PD) monitoring. If any internal damage is detected, the subcontractor is legally forced to replace the entire drum of cable at their own cost, and the specific installation crew is subjected to mandatory retraining before being allowed back on site.

Two subcontractors are in conflict over a work front. *Scenario:* The Civil subcontractor has fallen behind and has not finished curing the inverter pad. Simultaneously, the Electrical subcontractor has the central inverter suspended on a heavy-lift crane, waiting to be placed, and is accumulating hourly standby charges. *CM Response:* RAMPART acts as the sole arbitrator, driven entirely by the Master Schedule. Since the civil foundations have not reached the required compressive strength, the crane cannot place the load. The Electrical subcontractor is directed to demobilize the crane and pivot to an alternative work front (e.g., stringing DC modules). The demobilization and remobilization claims from the Electrical subcontractor are logged and subsequently back-charged against the underperforming Civil subcontractor in the next billing cycle.

Module delivery arrives at site, but the subcontractor is not ready to erect. *Scenario:* 50 MW worth of solar modules arrive at the site, but MMS erection is delayed, leaving nowhere to mount them. *CM Response:* The modules must be strategically staged in a designated, elevated, and secure laydown area. In Uzbekistan's arid climate, modules must be fiercely protected against intense dust storms. Accumulation of PM1.0 and PM2.5 dust particles causes rapid performance degradation, derating output by up to 30% if the glass is left uncleaned or if abrasive sand scratches the anti-reflective coating². The pallets are tightly strapped, covered with UV-resistant tarpaulins to prevent sand ingress into the cardboard packaging, and placed under 24/7 security watch to prevent theft of the project's most valuable assets.

An NCR is raised, but the subcontractor disputes it. *Scenario:* STRATUM issues an NCR because the tracker piles appear visually out of alignment and exceed the ± 20 mm East-West tolerance. The subcontractor's site manager disputes the measurement, claiming the EPC's tape measure is sagging. *CM Response:* RAMPART escalates the dispute strictly through data. The independent EPC surveyor (PRISM) is brought in to conduct a joint survey alongside the subcontractor's surveyor, utilizing a third, freshly calibrated Total Station. If the digital coordinates confirm the points remain out of tolerance, the NCR stands undeniably. If the subcontractor still refuses to rectify the piles, RAMPART leverages the contract: the EPC will hire a third-party contractor to pull and re-drive the piles, and the entire cost of this rectification is deducted from the original subcontractor's next Joint Measurement Certificate (JMC).

Works cited

1. GeoTech-III 2023, Geotech-IV 2024, Geotech-V 2024, Geotech-VI, <https://conf.domnit.ru/en/materials/geotech-2023/>
2. Sand and Dust Storms' Impact on the Efficiency of the Photovoltaic, <https://www.mdpi.com/1996-1073/16/9/3938>
3. The end of mass grading for solar projects - PV-Tech.org,

- <https://www.pv-tech.org/the-end-of-mass-grading-for-solar-projects/>
4. Field Test Study on the Bearing Capacity of Extra-Long PHC Pipe,
<https://www.mdpi.com/2071-1050/15/6/5161>
 5. How measuring soil resistivity helps optimize solar design,
<https://ratedpower.com/glossary/soil-resistivity/>
 6. Wenner Four-Pin Soil Resistivity Test | PDF | Electricity - Scribd,
<https://www.scribd.com/doc/191757451/Wenner-Four-Pin-Method-for-Soil-Resistivity-Test>
 7. Plant Tracker Installation Guide | PDF | Nut (Hardware) - Scribd,
[https://www.scribd.com/document/829823594/1-Installation-Manual-for-Tracker-Sy
stem-1P-En-KST00618-V1-1-1](https://www.scribd.com/document/829823594/1-Installation-Manual-for-Tracker-System-1P-En-KST00618-V1-1-1)
 8. INSTALLATION MANUAL OF STANDARD SOLAR MODULES,
[https://www.canadiansolar.com/wp-content/uploads/2020/08/Installation_Manual
of_Standard_Solar_Modules_en.pdf](https://www.canadiansolar.com/wp-content/uploads/2020/08/Installation_Manual_of_Standard_Solar_Modules_en.pdf)
 9. SF7 Tracker Assembly Guide | PDF | Photovoltaics | Solar Panel,
[https://www.scribd.com/document/742825136/MA-EN-00-en-Rev-1-Assembly-Ma
nual-SF7-Generic-Project-v01](https://www.scribd.com/document/742825136/MA-EN-00-en-Rev-1-Assembly-Manual-SF7-Generic-Project-v01)
 10. Custom Wire Cable Purchasing Guide for Global Buyers,
[https://www.pb-cable.com/blog-cable-connector-insights/custom-wire-cable-purch
asing-guide-for-global-buyers-12041370.html](https://www.pb-cable.com/blog-cable-connector-insights/custom-wire-cable-purchasing-guide-for-global-buyers-12041370.html)
 11. Solar PV Commissioning: IEC 62446 Tests Before Handover - MIMAH,
<https://mimah.net/posts/solar-pv-commissioning-testing>
 12. IEC 62446-1 - iTeh Standards,
[https://cdn.standards.iteh.ai/samples/20417/687a20e1bdfd42c4aef96348958655a
d/IEC-62446-1-2016.pdf](https://cdn.standards.iteh.ai/samples/20417/687a20e1bdfd42c4aef96348958655ad/IEC-62446-1-2016.pdf)
 13. PV Module IV Curve: A Definitive Guide to Understanding Solar,
[https://www.kippzonen.com/blog/what-is-a-pv-module-iv-curve-a-guide-to-underst
anding-solar-performance](https://www.kippzonen.com/blog/what-is-a-pv-module-iv-curve-a-guide-to-understanding-solar-performance)
 14. Heat Shrink vs Cold Shrink Cable Terminations Explained - Zoliov,
<https://www.zoliov.com/blog/heat-shrink-vs-cold-shrink-terminations/>
 15. MV Cable Accessory Manufacturers 2026 + Buyer Checklist - XBRELE,
<https://xbrele.com/mv-cable-accessory-manufacturers/>
 16. HV Cable Jointing & Termination GCC: Failure Modes & Best Practices,
<https://elecwatsgcc.com/blog/hv-cable-jointing-termination-gcc/>
 17. Medium Voltage Cable Termination: 10 Critical Errors to Avoid,
[https://www.centralwires.com/medium-voltage-termination-10-critical-errors-to-avoi
d/](https://www.centralwires.com/medium-voltage-termination-10-critical-errors-to-avoid/)
 18. 1 Hipot Cable Testing, <https://kep.ua/img/publications/files/hipot-cable-testing.pdf>
 19. Cable diagnostic in MV underground cable networks,
[https://www.baur.cz/wp-content/uploads/2019/11/BR_821-071_Cable_diagnostic_i
n_MV_underground_cable_network.pdf](https://www.baur.cz/wp-content/uploads/2019/11/BR_821-071_Cable_diagnostic_in_MV_underground_cable_network.pdf)
 20. HV Cable Testing - PDFCOFFEE.COM,
<https://pdfcoffee.com/hv-cable-testing-pdf-free.html>
 21. Testing procedures for high and low voltage power cables - Facebook,
<https://www.facebook.com/groups/3167634463285220/posts/2725091532786379>

[7/](#)

22. Understanding Power System Grounding: Principles, Guidelines,
<https://www.slideserve.com/alexandervazquez/power-system-grounding-powerpoint-ppt-presentation>
23. IEEE 80 Earthing System Calculation Guide | PDF - Scribd,
<https://www.scribd.com/document/625225788/Calculation-of-Earthing-System-as-Per-IEEE-80>
24. Transformer Installation Method Statement | PDF - Scribd,
<https://www.scribd.com/document/983170077/Transformer-Installation-Skidding-M-S>
25. Installation and Field Testing - Prolec Energy,
https://www.prolec.energy/wp-content/uploads/2026/05/Installation-and-Field-Testing_5.14.26-FNL-TK.pdf
26. Transformer Oil BDV (Breakdown Voltage) Testing Procedure,
<https://forumelectrical.com/transformer-oil-bdv-breakdown-voltage-testing-procedure/>
27. Bdv Test Of Transformer For Insulation Quality Assessment,
<https://www.huazhengtestequipment.com/bdv-test-of-transformer-for-insulation-quality-assessment/>
28. Transformer DGA Monitoring System: Complete Guide to Dissolved,
<https://www.fjinno.net/transformer-dga-monitoring-system>
29. Dissolved Gas Analysis: Transformer Fault Detection Guide,
<https://hd-transformer.com/blog/dissolved-gas-analysis-transformer/>
30. Transformer DGA Analyzer for Utilities: Complete Guide 2026,
<https://www.hvtesters.com/transformer-dga-analyzer-for-utilities-complete-guide-2026/>
31. SFRA testing on power transformers - Amon Energy,
<https://amonenergy.it/en/services/sfra/>
32. Sweep Frequency Response Analysis SFRA Training | PDF - Scribd,
<https://www.scribd.com/presentation/546534971/Training-2014-SFRA>
33. Cost-Effective SF6 Recycling and Purification for Grid Operators,
<https://www.sf6gasdetector.com/news/cost-effective-sf6-recycling-and-purification-for-grid-operators/>
34. IEC-60376-2018.pdf - iTeh Standards,
<https://cdn.standards.iteh.ai/samples/21595/aca5a27bf78490f9785a1ed268a9ba4/IEC-60376-2018.pdf>
35. What is the Maximum Allowed Water Content in SF6 for New Circuit,
<https://www.sf6gasdetector.com/news/what-is-the-maximum-allowed-water-content-in-sf6-for-new-circuit-breakers-per-neta-ats/>
36. Guide to Dry Type Transformer Maintenance - CHINT Global,
<https://www.chintglobal.com/sa/en/about-us/news-center/blog/dry-type-transformer-maintenance.html>
37. Power Room Shutdown and Re-energization Sequence - ALLPCB,
<https://www.allpcb.com/allelectrohub/power-room-shutdown-and-re-energization-sequence>

38. Analysing the switching sequence of a HV substation - Filo,
<https://askfilo.com/user-question-answers-smart-solutions/analysing-the-switching-sequence-of-a-hv-substation-3339383937363239>
39. PDO Construction/Commissioning Punch List | PDF - Scribd,
<https://fr.scribd.com/document/391557514/PUNCH>
40. IEC 61724, Solar Performance Monitoring Standard for Engineers,
<https://heavendesigns.in/blog/iec-61724-solar-performance-monitoring/>
41. Performance Ratio in Solar: Meaning and Why It Matters,
<https://www.heavengreenenergy.com/glossary/performance-ratio>
42. Project design > Results > Performance Ratio PR - PVsyst,
https://www.pvsyst.com/help-pvsyst7/performance_ratio.htm
43. The Effect of Dust Deposition on the Performance of Photovoltaic,
<https://www.mdpi.com/1996-1073/16/19/6794>